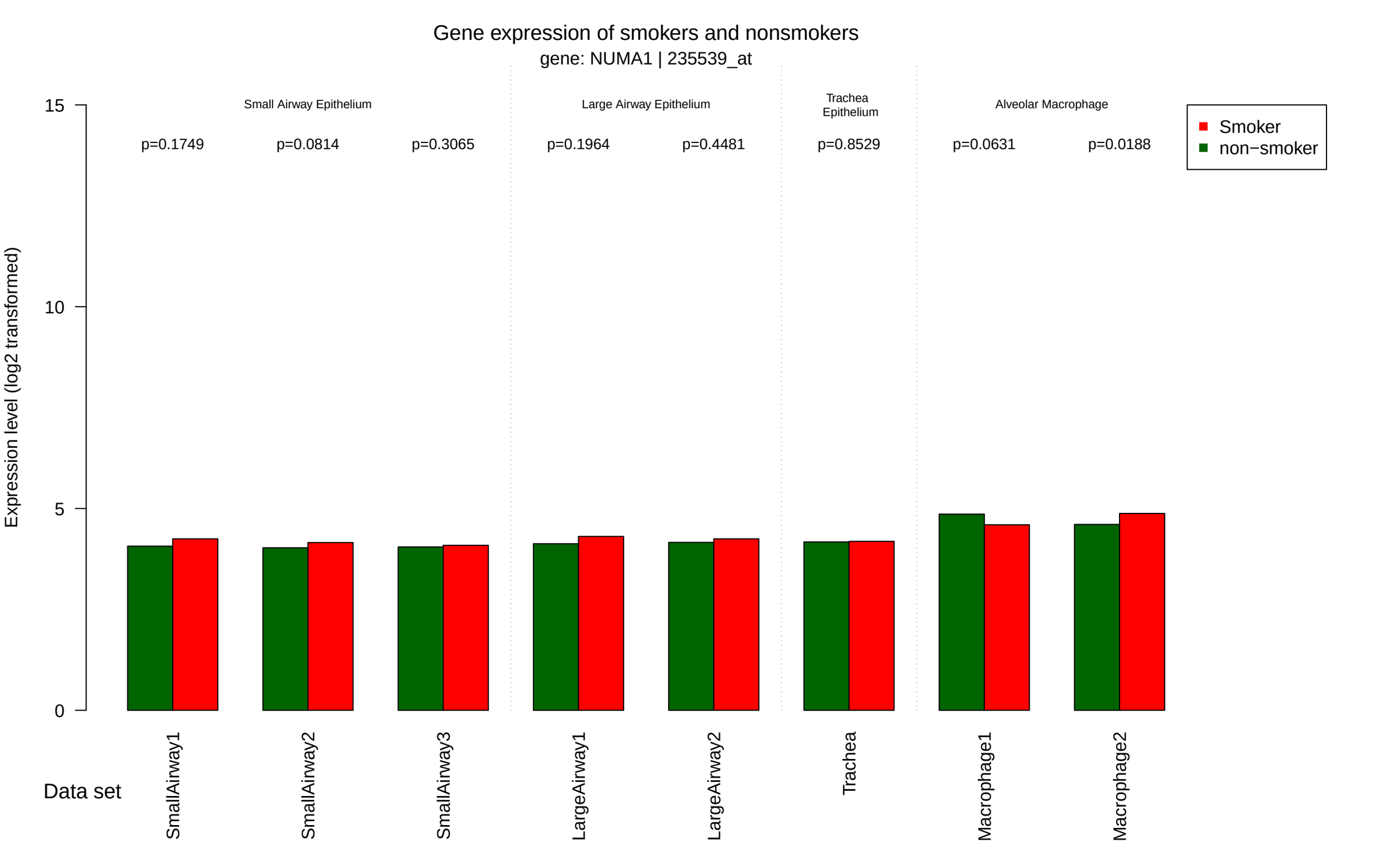
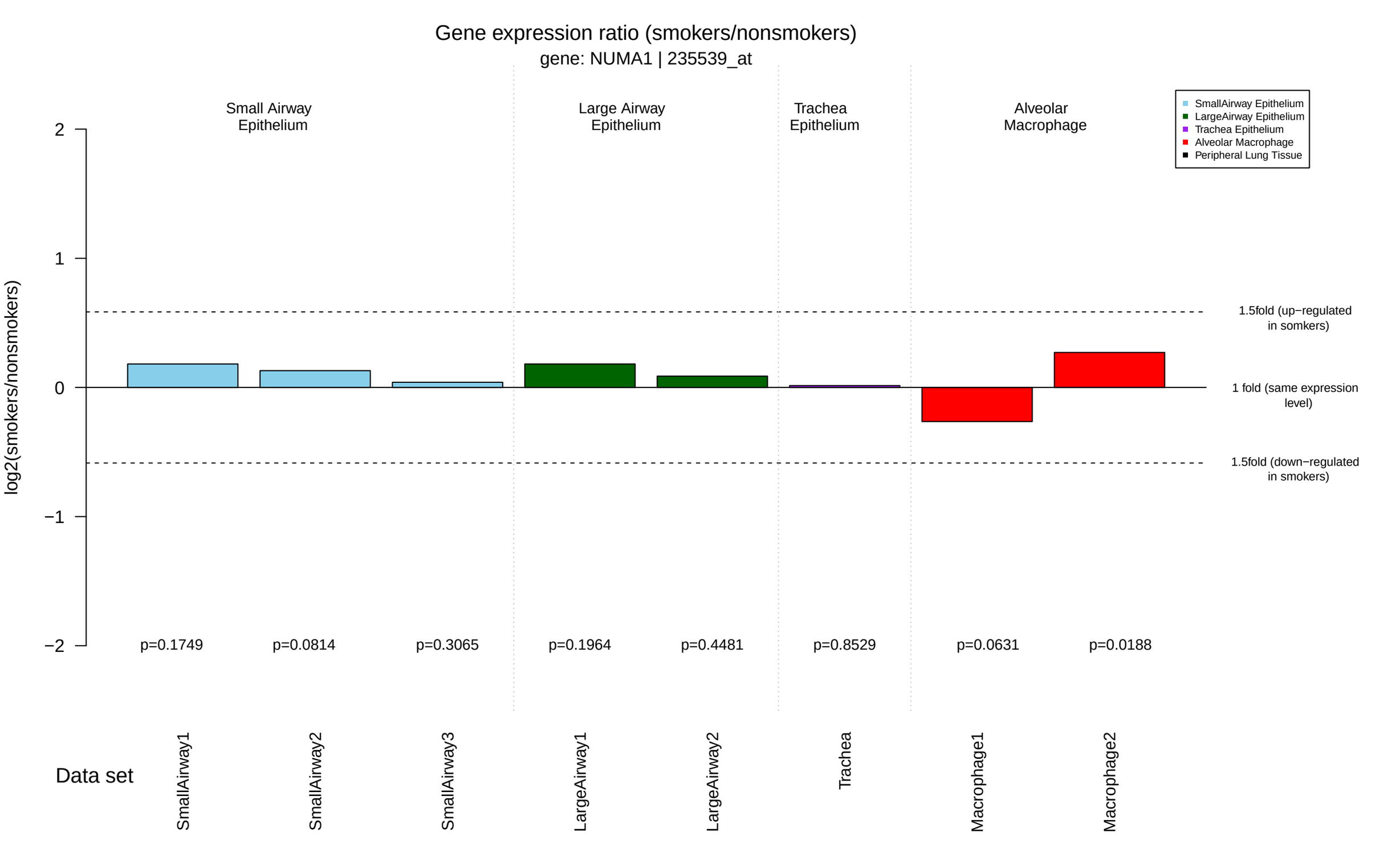
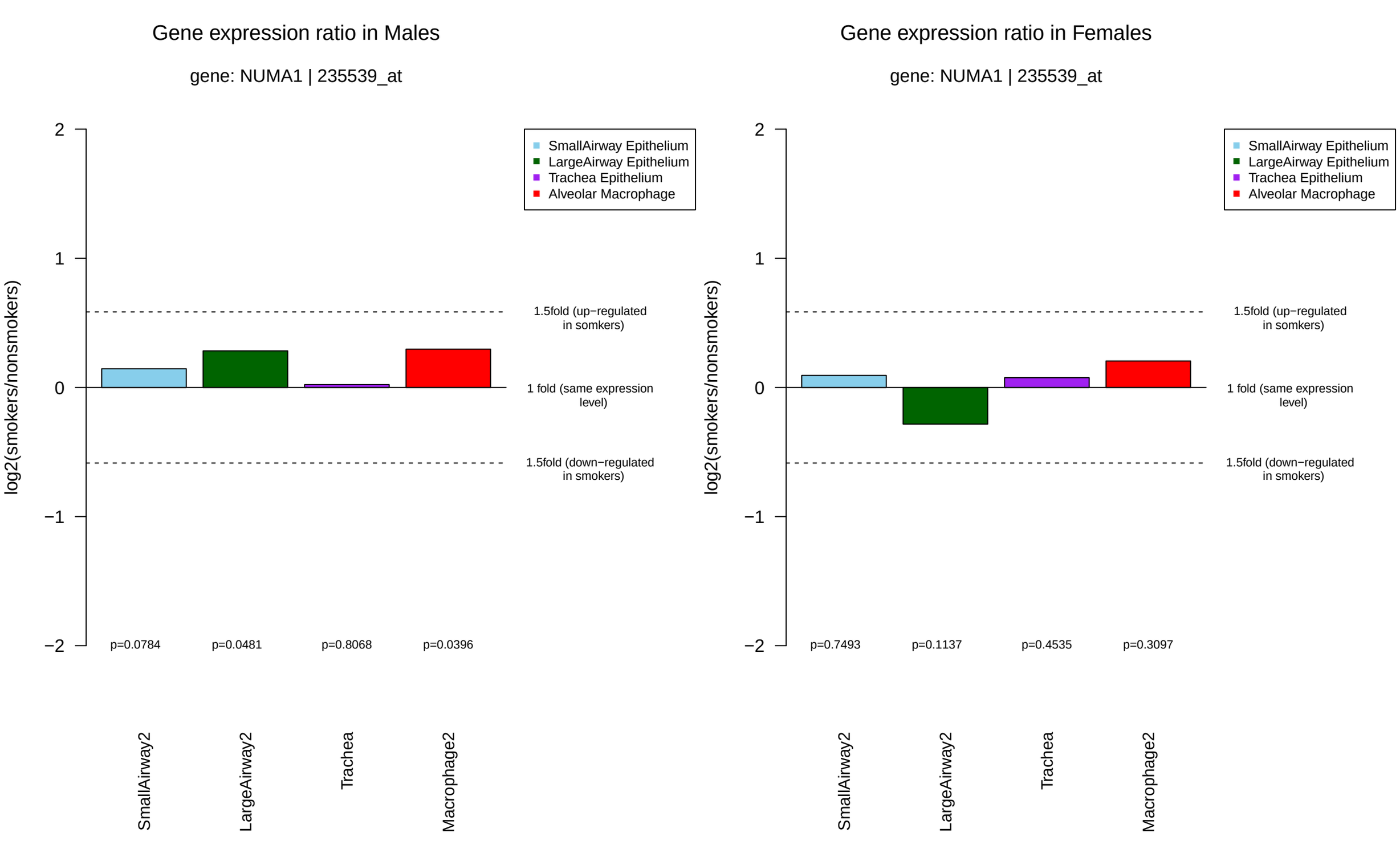
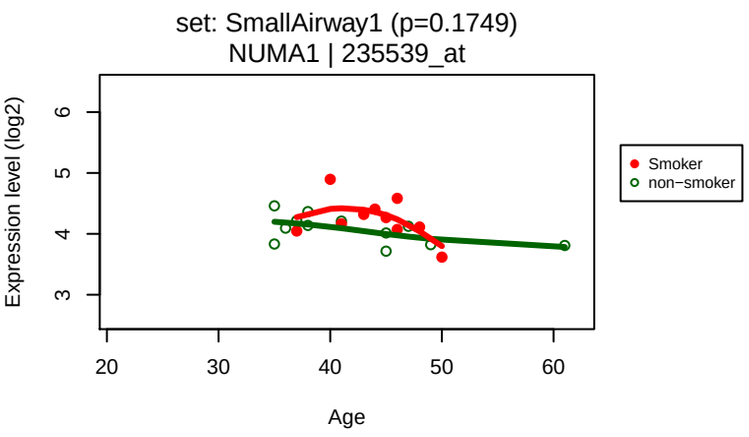
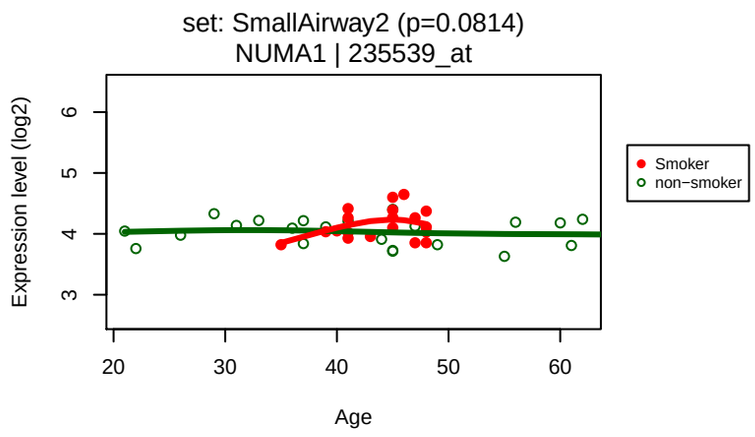
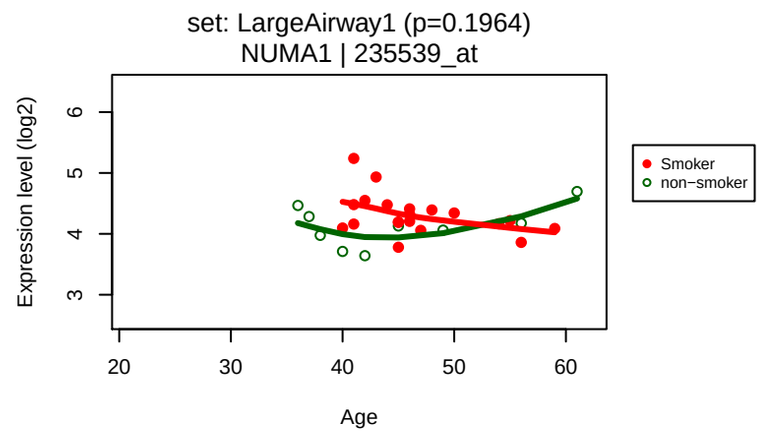

Gene expression of smokers and nonsmokers
gene: NUMA1 | 235539_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1749
p=0.0814
p=0.3065
p=0.1964
p=0.4481
p=0.8529
p=0.0631
p=0.0188
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

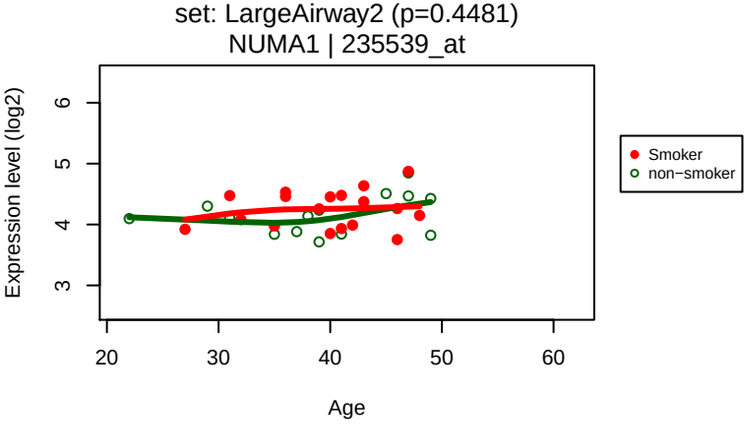

Gene expression ratio (smokers/nonsmokers)
gene: NUMA1 | 235539_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1749
p=0.0814
p=0.3065
p=0.1964
p=0.4481
p=0.8529
p=0.0631
p=0.0188
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

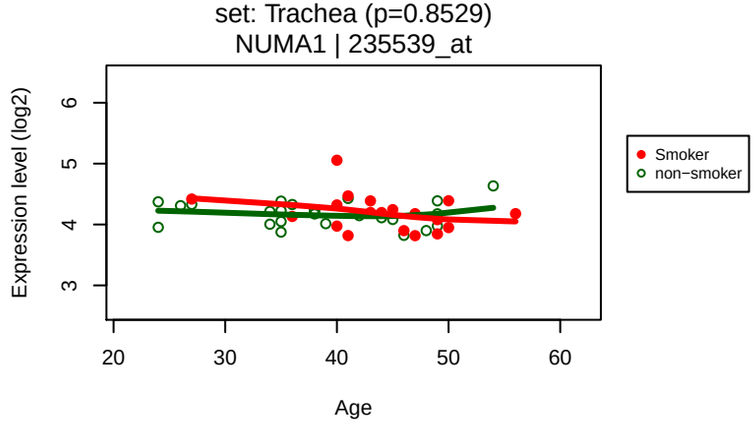

Gene expression ratio in Males
gene: NUMA1 | 235539_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0784
p=0.0481
p=0.8068
p=0.0396
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NUMA1 | 235539_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7493
p=0.1137
p=0.4535
p=0.3097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

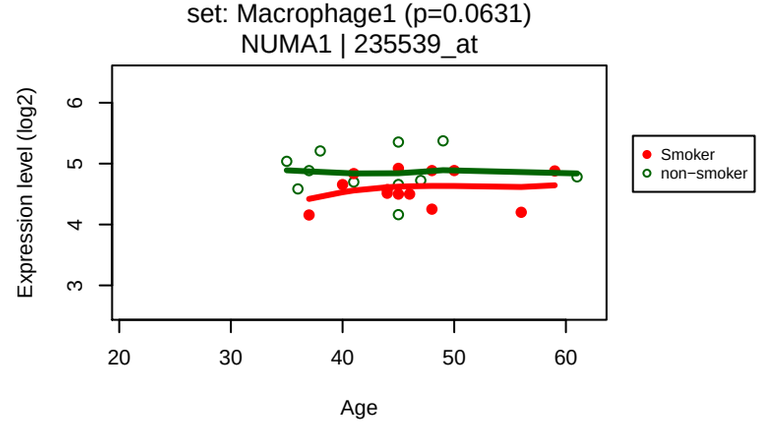

set: SmallAirway1 (p=0.1749)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker

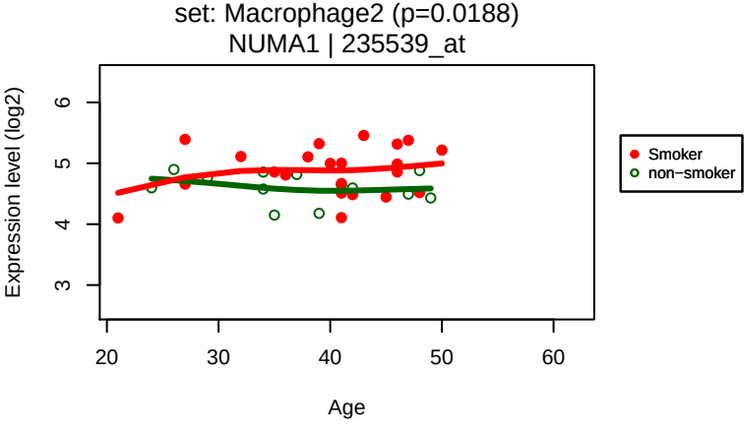

set: SmallAirway2 (p=0.0814)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker

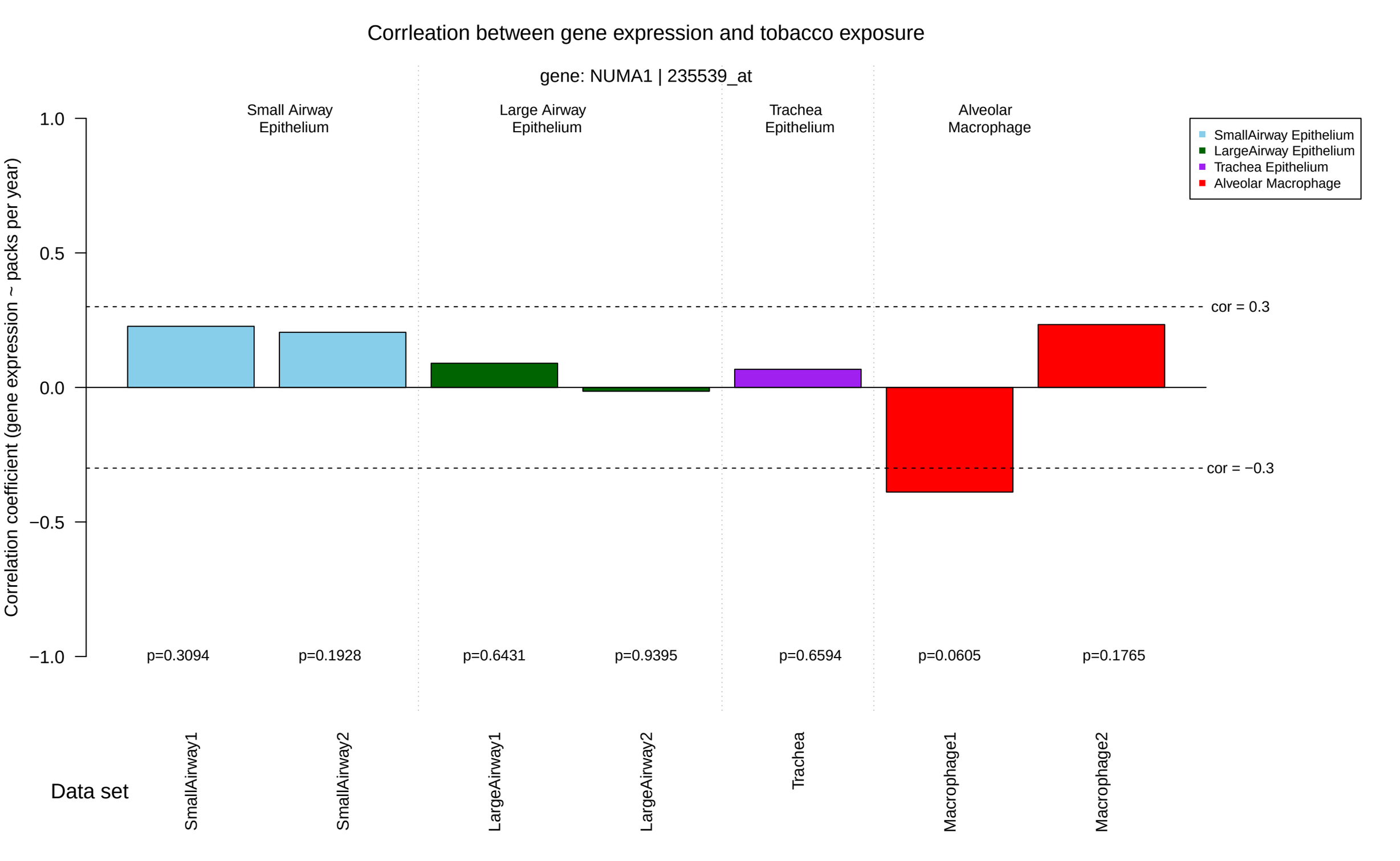

set: LargeAirway1 (p=0.1964)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4481)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8529)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0631)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0188)
NUMA1 | 235539_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NUMA1 | 235539_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3094
p=0.1928
p=0.6431
p=0.9395
p=0.6594
p=0.0605
p=0.1765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2